An independent exact recheck of a 2-tetrahedron counterexample to Luo's conjectures on Q-theory of closed 3-manifolds

(a recheck of a result by others; no novelty is claimed)

Xavier Callens (SocrateAI Lab)

Preprint v1, 2026-11-02

doi:10.5281/zenodo.23867965 (not peer reviewed)

Zenodo DOI to be assigned on release (not peer reviewed)

AI-assistance disclosure and audit status. This preprint, its code, its claim ledger and the audit were produced by AI agents (Claude, Anthropic) under the author's direction; the author is responsible for the content. **This version was audited by a separate AI agent instance (`AUDIT.md`, ACCEPT_WITH_CORRECTIONS), not by a human.** The corrections are applied (Section 8). No human audit exists. There is no Lean in this paper.

Abstract

Q. Guo [1] (arXiv:2610.13979, 11 Oct 2026, unrefereed; the paper says ChatGPT supplied the idea) exhibits a 2-tetrahedron, one-vertex triangulation T of the lens space $L(3,1)$ and claims that it refutes Conjectures 1–3 of F. Luo [2] (arXiv:1003.4413) on Haken's normal surface equation, Thurston's gluing equation and the maximum of the volume on the space of angle structures. **The result is Guo's; we claim nothing new,** and nothing about the Poincaré conjecture. Two independent implementations (the second with Regina and Singular as instruments) reproduce the four clauses of the counterexample: (i) $\pi_1 = C_3$ and a null-homotopic loop edge, (ii) Q-matching of rank 2 with no cluster of three 2-quad-type solutions, (iii) Thurston's equation has no solution (Groebner basis $\{1\}$), (iv) every maximum of the volume is a smooth point. Clause (iv) is upgraded from numerical to Tier [C] by an elementary analytic bound. Minimality of two tetrahedra is reproduced by a Regina census. Guo's quotation of Luo's conjectures was checked against arXiv:1003.4413v2. Outcome: confirmed as a recheck; Tier A (Lean) not reached.

What this is NOT. Not a new result; not a statement about the Poincaré conjecture or any simply connected case (Guo's Remark 1.2: $L(3,1)$ is not simply connected); not a formal proof (no Lean); not a comparison with the published Contemp. Math. 541 text of Luo; not interval arithmetic; not an exhaustive prior-art search; not human-audited.

Markers. [L]: a cited source, taken as stated. [C]: executed exact code (integer, rational, Groebner-basis arithmetic) or an elementary argument written down, not formalised. [X]: statements of what was not done and caveats. Identifiers `LUO-·-nnnn` refer to `ledger/ledger.json`.

1 The claim and its scope

Luo's conjectures concern a closed, irreducible, orientable, minimally triangulated 3-manifold M. Conjecture 1: either Thurston's equation (3.7) has a solution, or Haken's equation has a cluster of three 2-quad-type solutions. Conjecture 2: if all maxima of the volume on the space $\mathrm{SAS}(T)$ of (signed) angle structures are smooth points, then Thurston's equation has a solution. Conjecture 3: if some edge has equal endpoints and is null-homotopic, a cluster exists [2]. Guo's counterexample T has $\pi_1 = C_3$; clauses (ii)+(iii) refute C1, (iii)+(iv) refute C2, and (i)+(ii) refute C3 [1]. The deduction is elementary once the clauses and Luo's definitions are fixed. The auditor checked Guo's quotation of the conjectures and of the definitions ("2-quad-type", cluster, signed real Haken

solutions, Thurston's equation, SAS, smooth point) against Luo's arXiv v2 (pp. 2–3, 6–9, 12–14, 17–20) and found it faithful (`LUO-L-0002`). The Contemp. Math. 541 version was not compared. *Scope:* since $L(3,1)$ is not simply connected, the restricted versions behind Luo's route to the Poincaré conjecture are not touched (Guo's Remark 1.2).

2 Triangulation, homology, fundamental group ([C])

From the four face pairings the gluing permutations are $1230, 1230, 3201, 3201$, all odd; each of the 8 faces is glued once. There is one vertex class with link $(V, E, F) = (6, 12, 8)$, $\chi = 2$; three edge classes of degrees $2, 5, 5$; $\chi(M) = 0$; the local-edge table equals the paper's table (2). The isomorphism signature is `cMcabbjak`; Regina 7.4 finds the triangulation isomorphic to its lens$(3,1)$ and its census entry "$L(3,1) : \#3$". Smith normal forms of the auditor's own boundary matrices (d_2 of size 3×4, d_3 of size 4×2, $d_2 d_3 = 0$) give $H_1 = \mathbb{Z}/3$, $H_2 = 0$, $H_3 = \mathbb{Z}$. The fundamental group is C_3 by two routes: the producer's GAP computation and the auditor's sympy coset enumeration (order 3, and the subgroup $\langle a \rangle$ of index 3). Hence the loop edge a is null-homotopic (`LUO-C-0001`).

3 Q-matching and Lemma 3.1 ([C])

The matrices I and $W = I\Omega$ equal the paper's (5) entry by entry, and Regina's own quad matching matrix is identical to W. We have $\operatorname{rank} W = 2$ (kernel of dimension 4). The Haken system has rank 9 with a 5-dimensional solution space whose projection to the six quad coordinates equals $\ker W$. The only column pair of rank < 2 is (x_1, y_1), so every nonzero vector of $\ker W$ with at most two nonzero entries is a multiple of $(1, 0, 0, 1, 0, 0)$ (Lemma 3.1) and there is no cluster of three. The same holds for nonnegative, integer or signed solutions, since restricting only shrinks the set. A Regina vertex normal surface is a torus with quads $(1, 0, 0, 1, 0, 0)$, so a genuine 2-quad-type solution exists, but only in x_1, y_1 (`LUO-C-0002`).

4 Thurston's equation ([C])

The edge equations from the edge classes coincide with the paper's, with $w = 1/z$ the last two reduce to $1/z = 1$ and $z = 1$. The Groebner basis over $\mathbb{Q}$ (Singular, lex order, nondegeneracy $t\, z(1-z)w(1-w) = 1$) is $\{1\}$ for the edge products $(+, +, +)$, in both orientation conventions, so Thurston's equation has no solution. For the generalised equation with edge products in $\{\pm 1\}$, exactly $(+, -, -)$, $(-, +, -)$ and $(-, -, +)$ of the 8 patterns are solvable, in both conventions; $(+, -, -)$ gives $z = w = -1$ (Guo's Remark 4.2). These solvable patterns serve as positive controls (`LUO-C-0003`).

5 Clause (iv): the volume maximum ([C])

Numerical by the producer: on SAS(T) (rank-4 constraints, a two-parameter family) a grid search with Nelder–Mead refinement gives the maximum 1.58864663930016 at 8 symmetric smooth points; the auditor reproduces it as a strict local maximum (Hessian eigenvalues $-13.7, -50.4$, $\min |\sin\theta| = 0.3928$).

Analytic bound by the auditor (upgrade to [C]). Angles are taken mod π; the Lobachevsky function Λ is odd and π-periodic. In a tetrahedron the three angles sum to $\pi \bmod 2\pi$; if one is in $\pi\mathbb{Z}$, the other two sum to 0 mod π and their Λ values cancel, so a partially flat tetrahedron contributes 0 volume. Since $\Lambda'(\theta) = -\log|2\sin\theta|$, $|\Lambda| \le \Lambda(\pi/6)$ everywhere, and $\Lambda(\pi/6) \le 0.507721$ by a partial sum with tail bound $1/(2N)$. At a non-smooth point one of the two tetrahedra is partially flat, so $\mathrm{vol} \le 3\Lambda(\pi/6) \le 1.52316$. On the other hand the exact rational point $\theta/\pi = (3/5, 1/8, 11/40, 7/5, 17/40, 47/40)$ lies in SAS(T) (tetrahedron sums 1 and 3, edge sums $2, 4, 2$, checked in rationals), all six angles are non-integer multiples of π, and its volume is at least 1.58529 (partial sum $N = 2000$ of $\Lambda(\theta) = \frac{1}{2}\sum \sin(2n\theta)/n^2$ with tail bound 0.0015). Since $1.58529 > 1.52316$, every maximum is smooth. The numbers are 30-digit floats with a rigorous tail bound, not interval arithmetic; the rounding error is about 10^{-25}, far below the margin 0.062. This route uses neither Luo's Theorems 4.2(a), 4.4 nor Milnor's bound.

Side remark. Using Milnor's bound ([L]), the non-smooth supremum is exactly $v_3 = 3\Lambda(\pi/3) = 1.0149416$, attained at $\theta/\pi = (2/3, 0, 1/3, -2/3, 1/3, 4/3)$ (exact membership in SAS checked). The producer's sampled non-smooth value 1.0149 is this supremum, not a sampling artefact (`LUO-C-0004`).

6 Minimality ([C])

By Regina enumeration, none of the 108 one-tetrahedron gluings has $H_1 = \mathbb{Z}/3$, so two tetrahedra is minimal (this reproduces, rather than cites, the Jaco–Rubinstein–Tillmann fact). Among the 136080 two-tetrahedron gluings, 5088 are closed, orientable and valid, in 16 classes; three have $H_1 = \mathbb{Z}/3$ and $\pi_1 = \langle a \mid a^3 \rangle$: `cMcabbgaj` (one vertex, degrees $1, 2, 9$), `cMcabbjak` (Guo's) and `cPcbbbaai` (two vertices). **So Guo's triangulation is not the only one-vertex minimal triangulation of $L(3, 1)$.** This does not affect the counterexample, but the other one-vertex triangulation was not tested against C1–C3. Calling the other classes "$L(3, 1)$" uses geometrization ([L]); for Guo's own triangulation Regina's explicit lens space construction suffices (`LUO-C-0005`).

7 Controls

Positive: Regina's figure-eight knot complement triangulation is solvable with $z = w = e^{\pm i\pi/3}$, and SnapPy's 4_1 shapes satisfy the edge equations to $1.4 \cdot 10^{-15}$ (this does not distinguish the two orientation conventions); the producer's π_1 code gives C_3, C_5, C_7, C_2 and trivial for $L(3, 1), L(5, 2), L(7, 2), L(2, 1), S^3$, matching Regina. Negative: 0 of 20 single-pairing perturbations keep the claim profile (one vertex, degrees $2, 5, 5$, $|\pi_1| = 3$, same I); 18 single-entry increments of I are all unsolvable, a weak control because the solver never said yes there; the discriminating "yes" cases are the figure-eight and the sign patterns of Section 4 (`LUO-C-0006`).

8 Audit corrections applied

The producer's `data/RESULT.md` and `data/result.json` are kept unmodified (pre-audit); this paper supersedes them on the audit's five corrections. (1) Orientation: in the producer's presentation b, c are oriented opposite to the paper's convention; with the paper's orientation the relators are exactly the paper's (3). (2) Clause (iv) is upgraded from numerical to Tier [C] (Section 5), and 1.0149 is identified as the exact non-smooth supremum v_3. (3) Luo's definitions and conjectures are checked against arXiv:1003.4413v2, and Guo's Remark 1.2 scope is added; the published Contemp. Math. text was not compared. (4) Minimality is reproduced and the non-uniqueness of the one-vertex triangulation is stated. (5) Outcome: (i)–(iii) [C], (iv) [C], minimality [C], Tier A not reached. The preregistration (`PREREG_ITEM.md`, sha256 `106565b6...4dca57`) is self-attested; no independent timestamp.

9 Limitations and what is not done

- AI audit only; no human audit; no Lean (no Tier A).
- Not verified: Milnor's v_3 bound (side remark only); the step "$\pi_1 = \mathbb{Z}/3 \Rightarrow L(3, 1)$" (geometrization) used to name the other census classes; the Contemp. Math. 541 text of Luo; Segerman–Tillmann (Guo's Remark 4.3); interval arithmetic for the volume bounds.
- The other one-vertex minimal triangulation (`cMcabbgaj`) was not tested against C1–C3.
- Literature: no independent verification or critique of Guo's paper and no earlier refutation of Luo's conjectures was found, but the search was limited (`LUO-L-0003`); "not found" is relative to the queries.
- Trusted components: Regina 7.4, Singular, GAP 4.11.1, sympy, mpmath, SnapPy.

10 Claim ledger and reproducibility

`ledger/build_ledger.py` files the claims with hashed evidence; the Elenchus gate (`ledger/run_gate.py`, verbatim output in `ledger/gate_output.txt`) checks bookkeeping only and licenses nothing. Producer code

and summaries are in `data/code`, `data/summaries`; the auditor's scripts (`a1`–`a5`, `thurston.sing.txt`) and small outputs are in `data/audit`. The papers' full texts are third-party and not shipped. Nothing was committed, pushed or uploaded.

References

[1] Q. Guo, *A Counterexample to Three Conjectures of Luo*, arXiv:2610.13979 (v1, 11 Oct 2026), unrefereed. ([L])

[2] F. Luo, *Triangulated 3-Manifolds: from Haken's normal surfaces to Thurston's algebraic equation*, arXiv:1003.4413 (v1 23 Mar 2010; v2 23 Jul 2010). ([L])